Fitness analysis: seed FW and germination rate

Table of contents

1 Setup **1**
- 1.1 library . . . 1
- 1.2 Data . . . 1

2 Aphid effect on seed FW **2**

3 Aphid effect on germination rate **4**
- 3.1 Plot of germination rate = f(genotype, aphid treatment) . . . 4
- 3.2 Statistical tests germination rate = f(genotype, aphid treatment) . . . 5

1 Setup

Analysis of the fitness of selected *Vicia faba* genotypes infested or not with *A. fabae* AF001 or *A. pisum* P123amp aphids. Here, we focus on seed fresh weight and germination rate.

See the data source.

1.1 library

1.2 Data

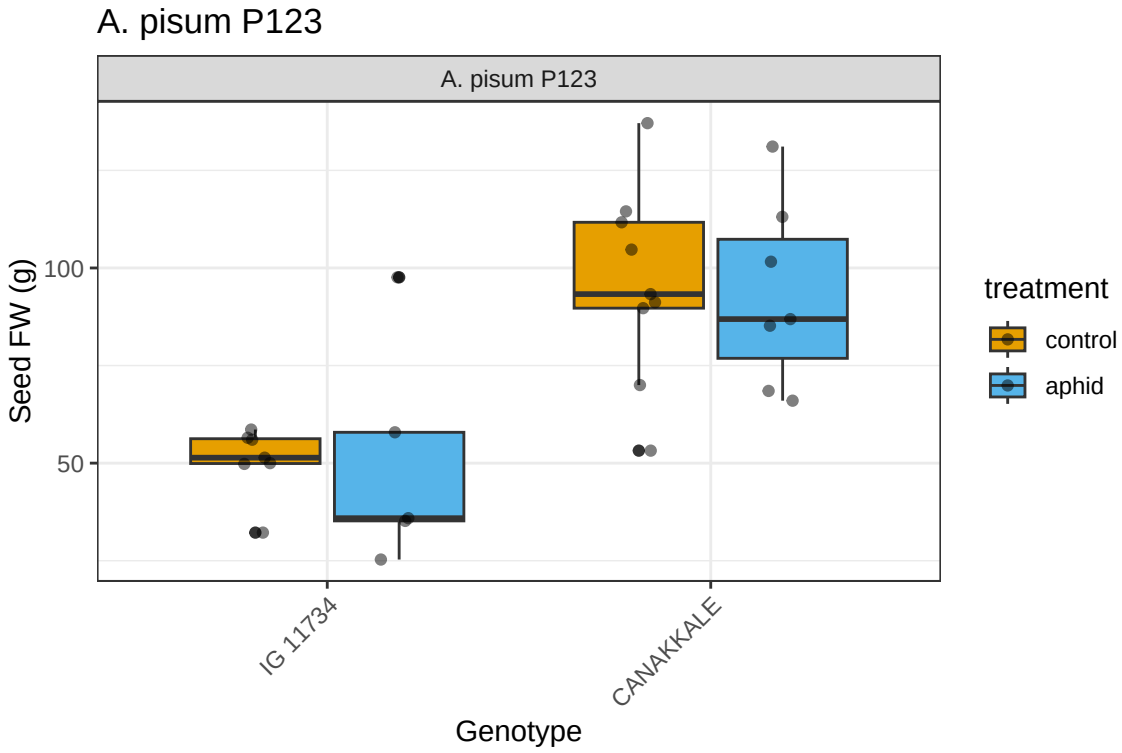
```
data_fitness %>%
  group_by(name, treatment, aphid_species) %>%
  summarise(n_values = n())
```

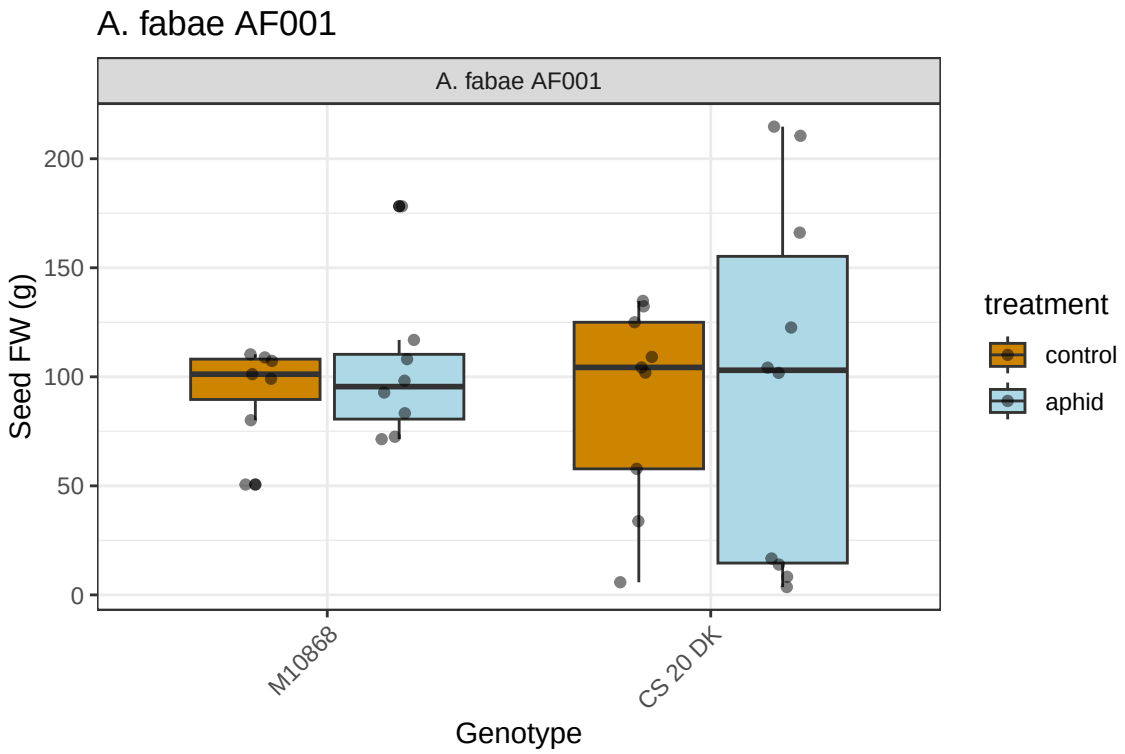
```
`summarise()` has grouped output by 'name', 'treatment'. You can override using
the `.groups` argument.

# A tibble: 8 x 4
# Groups:   name, treatment [8]
  name      treatment aphid_species  n_values
  <fct>     <fct>     <chr>             <int>
1 IG 11734  control   A. pisum P123        10
2 IG 11734  aphid     A. pisum P123        10
3 CANAKKALE control   A. pisum P123        10
4 CANAKKALE aphid     A. pisum P123        10
5 M10868    control   A. fabae AF001       10
6 M10868    aphid     A. fabae AF001       10
7 CS 20 DK  control   A. fabae AF001       10
8 CS 20 DK  aphid     A. fabae AF001       10
```

2 Aphid effect on seed FW

2.0.1 Plots of FW seeds = f(genotype, aphid treatment)





2.0.2 Statistical tests FW seed = f(genotype, aphid treatment)

ANOVA for the effect of A. pisum P123 on the seed FW of pods.

term	df	sumsq	meansq	statistic	p.value	significance
treatment	1	5.375744	5.375744	0.0106526	0.9186529	ns
name	1	13491.872523	13491.872523	26.7354896	0.0000270	***
treatment:name	1	12.148653	12.148653	0.0240738	0.8779949	ns
Residuals	24	12111.427365	504.642807	NA	NA	ns

ANOVA for the effect of A. fabae AF001 on the seed FW

term	df	sumsq	meansq	statistic	p.value	significance
treatment	1	503.036802	503.036802	0.1670655	0.6856363	ns
name	1	256.234479	256.234479	0.0850990	0.7725105	ns
treatment:name	1	7.766055	7.766055	0.0025792	0.9598327	ns
Residuals	30	90330.488841	3011.016295	NA	NA	ns

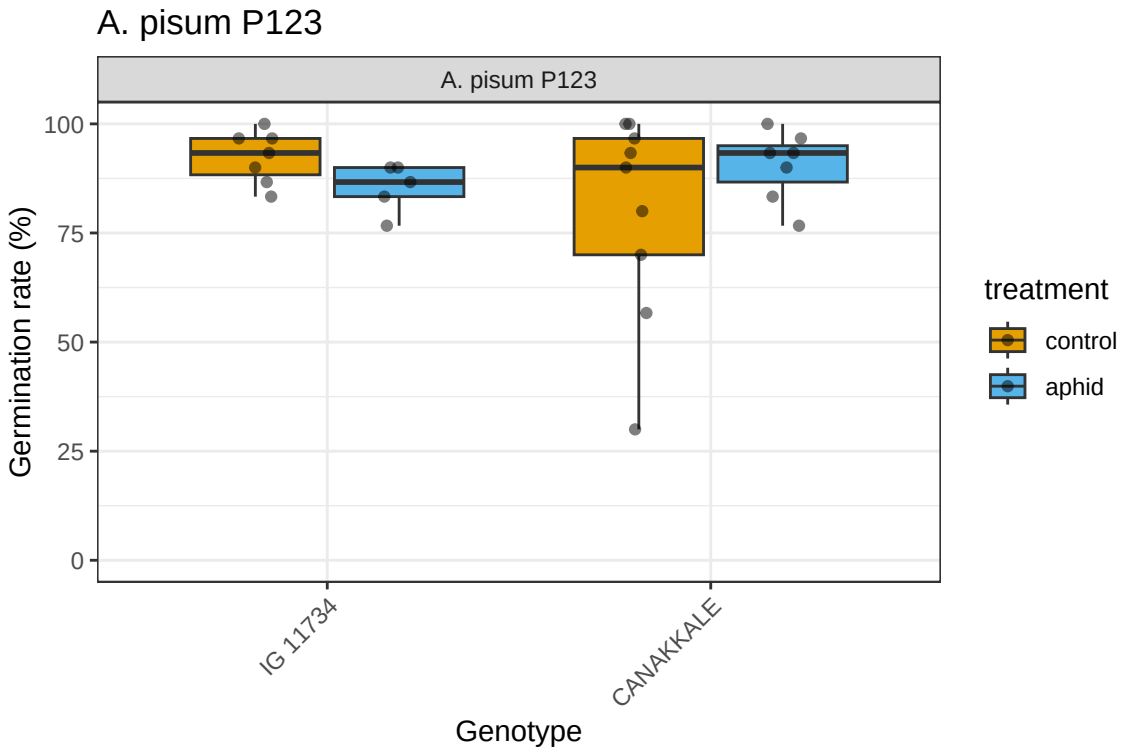
! Important

For *A. pisum* P123 and *A. fabae* AF001:
No effect of aphid treatment on the seed FW. Variability is high between plants and the seed FW is predominantly affected by the plant genotype.

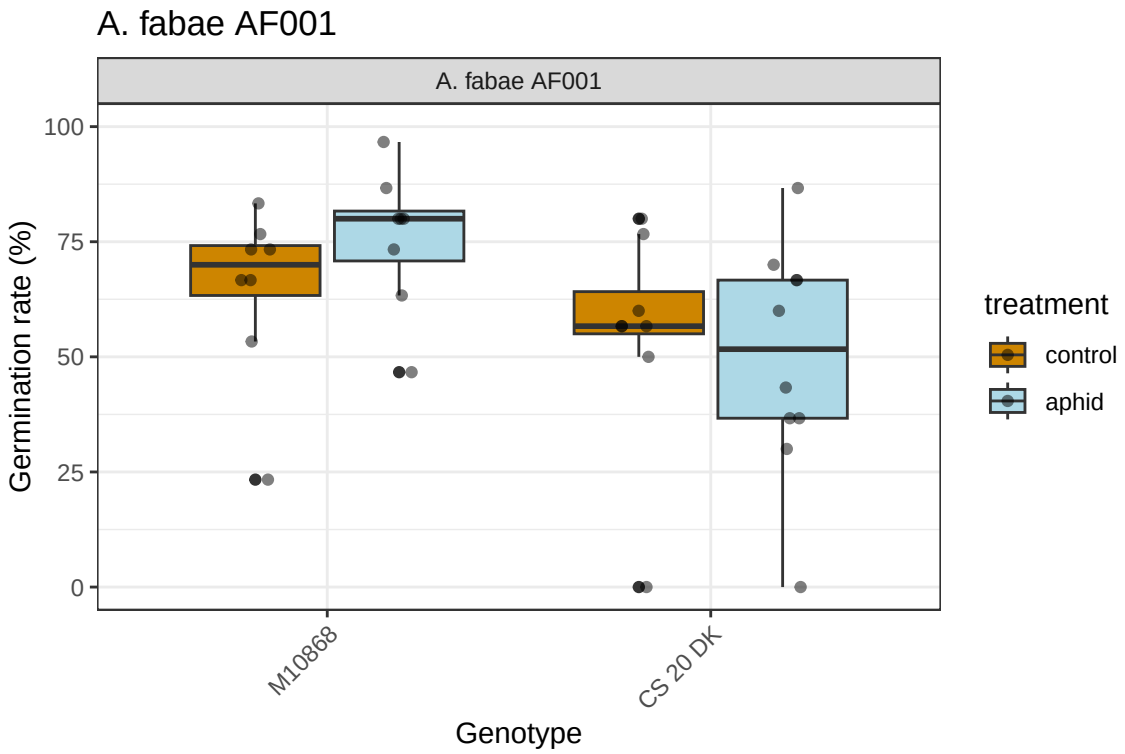
3 Aphid effect on germination rate

3.1 Plot of germination rate = f(genotype, aphid treatment)

A. pisum P123



A. fabae AF001



3.2 Statistical tests germination rate = f(genotype, aphid treatment)

ANOVA for the effect of A. pisum P123 on the germination rate of seeds.

term	df	sumsq	meansq	statistic	p.value	significance
treatment	1	66.96429	66.96429	0.3075057	0.5843460	ns
name	1	180.85740	180.85740	0.8305125	0.3711876	ns
treatment:name	1	536.50812	536.50812	2.4636907	0.1295964	ns
Residuals	24	5226.38448	217.76602	NA	NA	ns

ANOVA for the effect of A. fabae AF001 on the germination rate of seeds.

term	df	sumsq	meansq	statistic	p.value	significance
treatment	1	24.85476	24.85476	0.0533385	0.8189205	ns
name	1	2892.85250	2892.85250	6.2080847	0.0184725	*
treatment:name	1	550.23392	550.23392	1.1808064	0.2858461	ns
Residuals	30	13979.44444	465.98148	NA	NA	ns